CNENGR202 Instrumentation and Control

Part 1 L5 – Time & Frequency Analysis

First order systems

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Dynamic Response of Systems

$$\tau \frac{dx}{dt} + x = Ku(t)$$

$$\tau \dot{x} + x = Ku(t)$$

$$\frac{d^2x}{dt^2} + 2\zeta\omega_n \frac{dx}{dt} + \omega_n^2 x = Ku(t)$$

$$\ddot{x} + 2\zeta\omega_n \dot{x} + \omega_n^2 x = Ku(t)$$

Outline

- **Introduction**
- **First order systems**
  - Solution
  - Standard inputs
  - Time response
  - Frequency response
- **Second** order systems
  - Solution
  - Time response
  - Frequency response

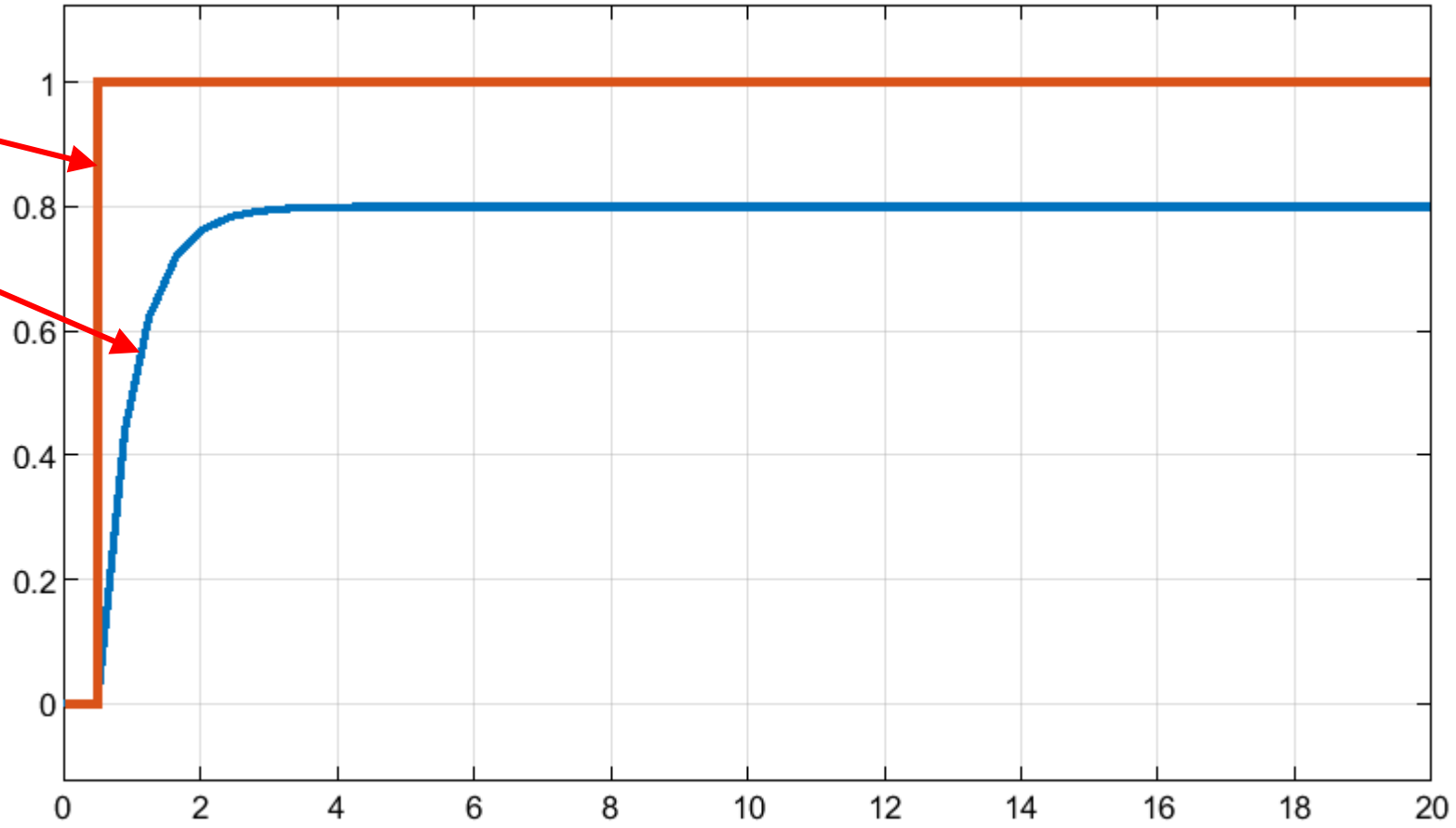
Introduction - 1

- A wide range of physical systems can be represented by linear 1st or 2nd order differential equations
- These models allow us to analyse and make predictions about the behaviour of the system before it is actually built
- Learning Objectives. At the end of this section, you should be able to:
  - Define common input signals such as the impulse, step, ramp and harmonic input
  - Understand how the model coefficients influence both the time and frequency response of 1st and 2nd order systems
  - Estimate the coefficients of these models from time and frequency response data

Introduction - 2

1st Order Simulation Example

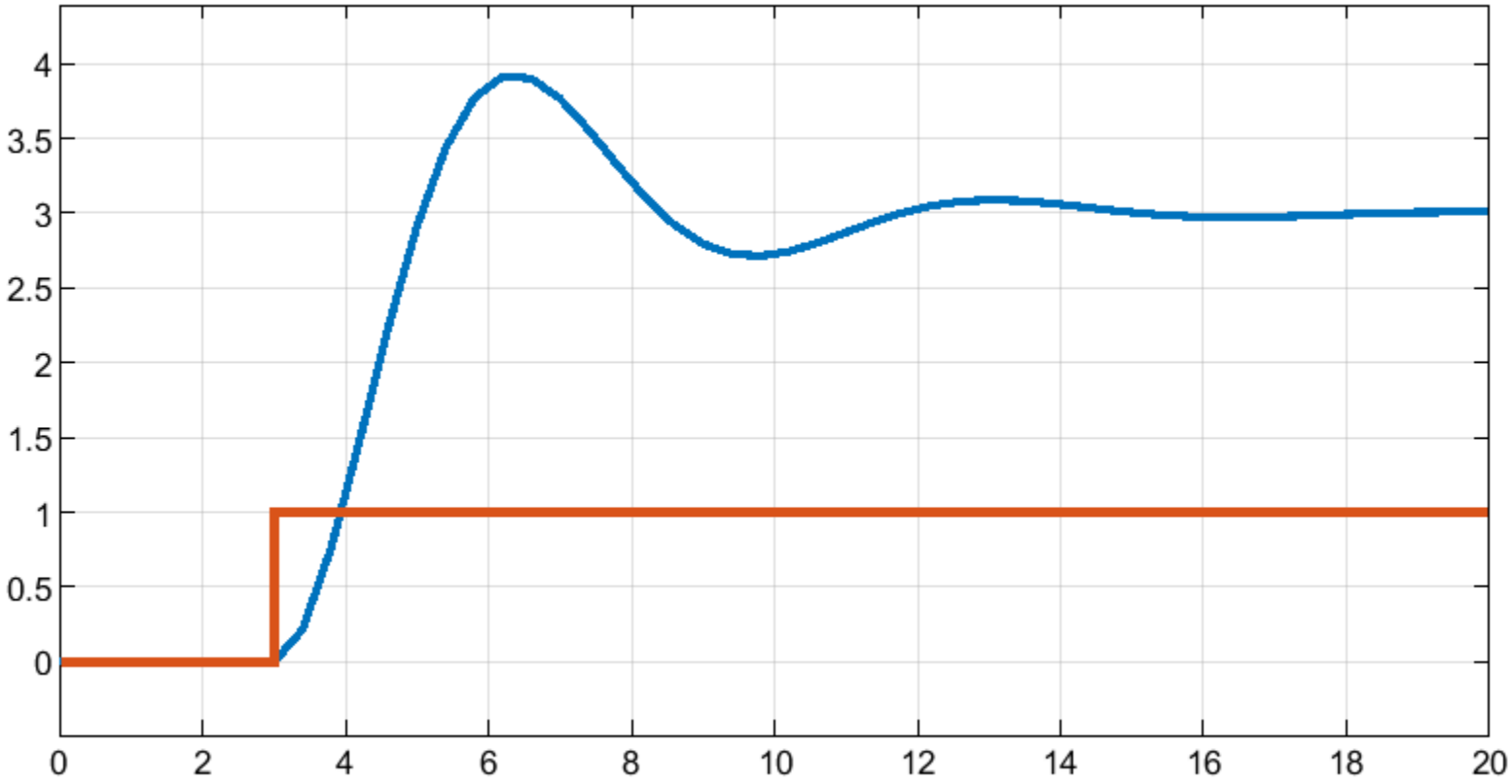
$$\tau \frac{dx}{dt} + x = Ku(t)$$



Introduction - 3

$$\frac{d^2x}{dt^2} + 2\zeta\omega_n \frac{dx}{dt} + \omega_n^2 x = Ku(t-3)$$

2nd Order Simulation Example



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1st Order - Solution - 1

$$\tau \frac{dx}{dt} + x = Ku$$

- Laplace transformation $\tau sX(s) + X(s) = \frac{K}{s}$
- $X(s)$ $X(s) = \frac{K}{s(\tau s + 1)} = K\left(\frac{1}{s} - \frac{1}{s + \frac{1}{\tau}}\right)$

1st Order - Solution - 2

- $X(s)$

$$X(s) = \frac{K}{s(\tau s + 1)} = K\left(\frac{1}{s} - \frac{1}{s + \frac{1}{\tau}}\right)$$

- $x(t)$

$$x(t) = K\left(1 - e^{-t/\tau}\right)$$

Example 1

$$X(s) = \frac{4s + 5}{s^2 + 5s + 6}$$

Solution for Example 1

$$X(s)=\frac{4s+5}{s^2+5s+6}=\frac{4s+5}{(s+2)(s+3)}=-\frac{3}{s+2}+\frac{7}{s+3}$$

$$x(t)=-3e^{-2t}+7e^{-3t}$$

Example 2

$$X(s) = \frac{2s+1}{s^3+7s^2+10s}$$

Solution for Example 2

$$X(s)=\frac{2s+1}{s^3+7s^2+10s}=\frac{2s+1}{s(s+2)(s+5)}=\frac{0.1}{s}+\frac{0.5}{s+2}-\frac{0.6}{s+5}$$

$$x(t)=0.1+0.5e^{-2t}-0.6e^{-5t}$$

Outline

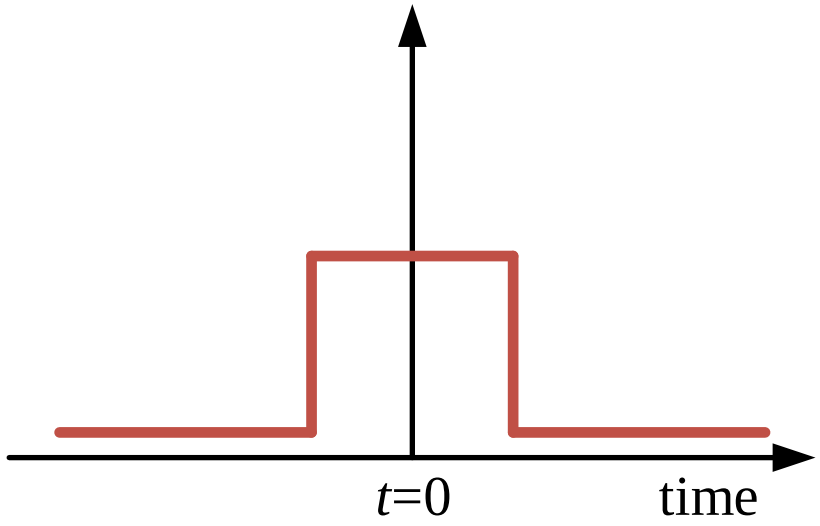
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Impulse -1

- Also called the Dirac Delta function

$$\delta(t) = 0 \qquad t \neq 0$$

$$\int_{-\infty}^{\infty} \delta(t)\,dt = 1$$

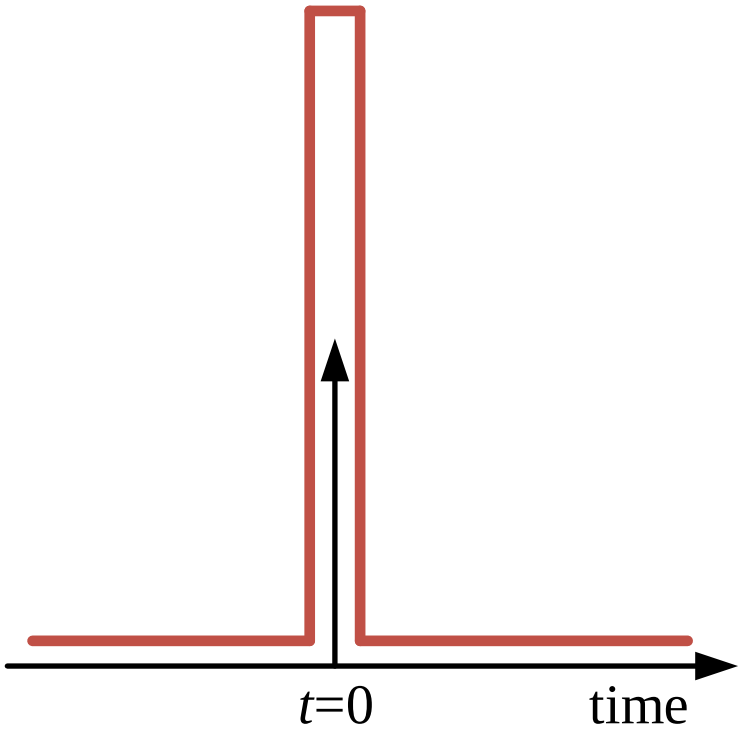


Impulse -2

- Also called the Dirac Delta function

$$\delta(t) = 0 \qquad t \neq 0$$

$$\int_{-\infty}^{\infty} \delta(t)\,dt = 1$$



Unit Step

$$u(t)=\begin{cases}0 & \text{for} \quad t<0\\ 1 & \text{for} \quad t\geq 0\end{cases}$$

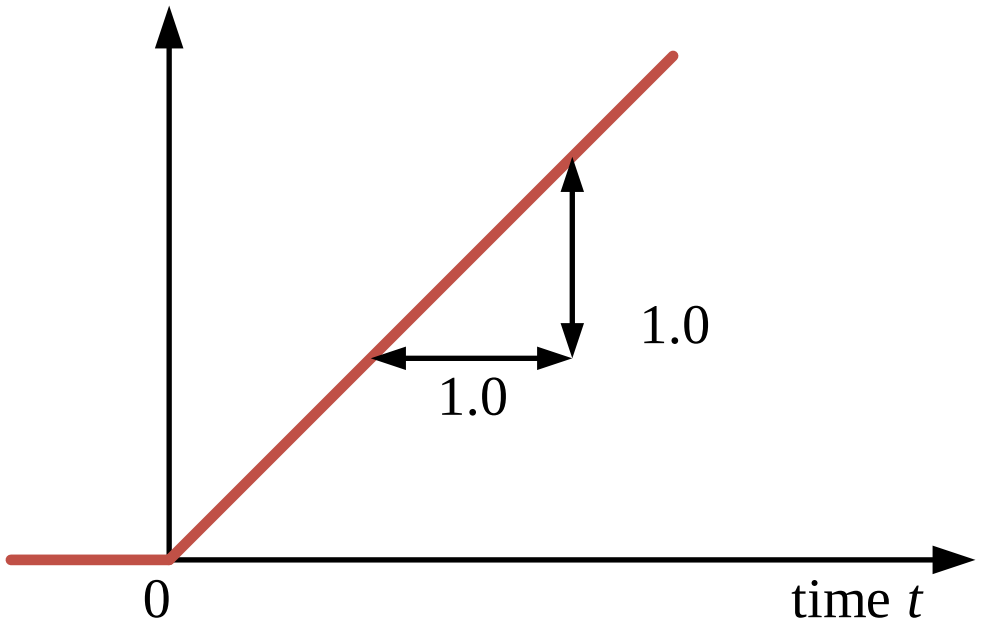
$$u(t)=\int_{-\infty}^{t}\delta(t)\,dt$$



Unit Ramp

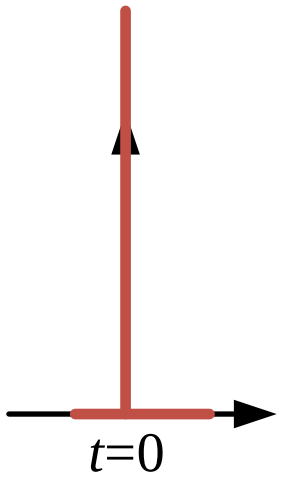
$$r(t) = \begin{cases} 0 & \text{for} \quad t < 0 \\ t & \text{for} \quad t \geq 0 \end{cases}$$

$$r(t) = \int_{-\infty}^{t} u(t)\,dt$$



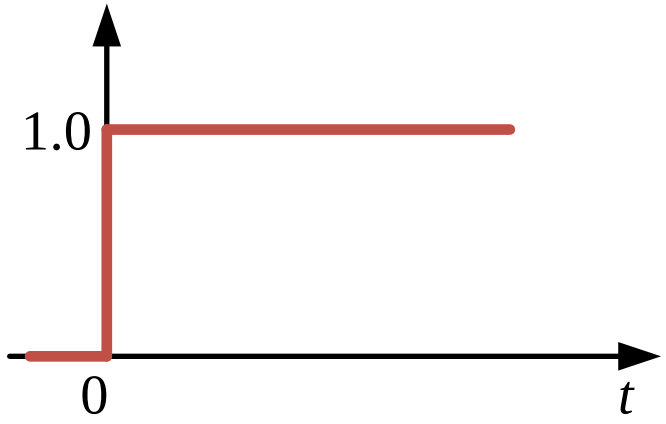
1st Order – Standard Inputs

Integrate



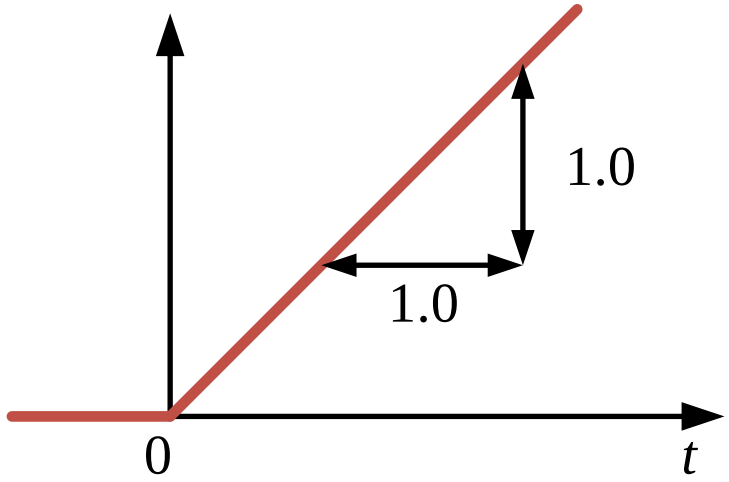
Impulse

$\delta(t)$



Step

$u(t)$



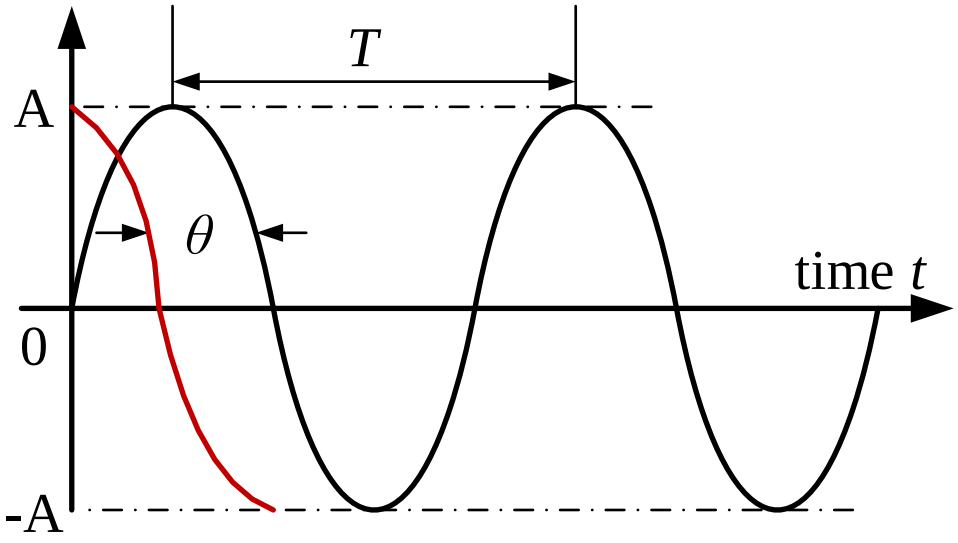
Ramp

$r(t)$

Differentiate

Harmonic Inputs

- Cosine Function $u(t) = A\cos(2\pi ft + \theta)$
- Amplitude A
- Angular frequency (rads/s) $\omega = 2\pi f$
- Frequency f (Hz)
- Period $T = 1/f$ (s)
- Phase θ (rads)



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Response to a Impulse -1

- 1st order differential equation $$\tau \frac{dx}{dt} + x = K\delta$$
- Laplace transformation $$X(s) = \frac{K}{\tau s + 1}$$
- General solution $$x(t) = \frac{K}{\tau} e^{-t/\tau}$$

Response to a Impulse - 2

- Steady State Gain

$$x(\infty) = \frac{K}{\tau} e^{-\infty/\tau} = 0$$

$$x(\infty) = \lim_{s \to 0} sX(s) = \lim_{s \to 0} \frac{sK}{\tau s + 1} = 0$$

- Initial Conditions

$$x(0) = \frac{K}{\tau} e^{-0} = \frac{K}{\tau}$$

$$x(0) = \lim_{s \to \infty} sX(s) = \lim_{s \to \infty} \frac{sK}{\tau s + 1} = \frac{K}{\tau}$$

Example3

if the time constant is 2s and K=1

$$\tau \frac{dx}{dt} + x = K\delta \qquad \Rightarrow \qquad x(t) = ?$$

Solution for example3

if the time constant is 2s

τ=2

$$\tau \frac{dx}{dt} + x = K\delta \qquad \Rightarrow \qquad 2\frac{dx}{dt} + x = \delta$$

$$x(t) = \frac{K}{\tau} e^{-t/\tau} \qquad \Rightarrow \qquad x(t) = \frac{1}{2} e^{-t/2}$$

Response to a Unit Step - 1

$$\tau\frac{dx}{dt}+x=Ku(t)$$

$$X(s)=\frac{K}{s(\tau s+1)}=K\left(\frac{1}{s}-\frac{1}{s+1/\tau}\right)$$

$$x(t)=K\left(1-e^{-t/\tau}\right)$$

- Steady State Gain

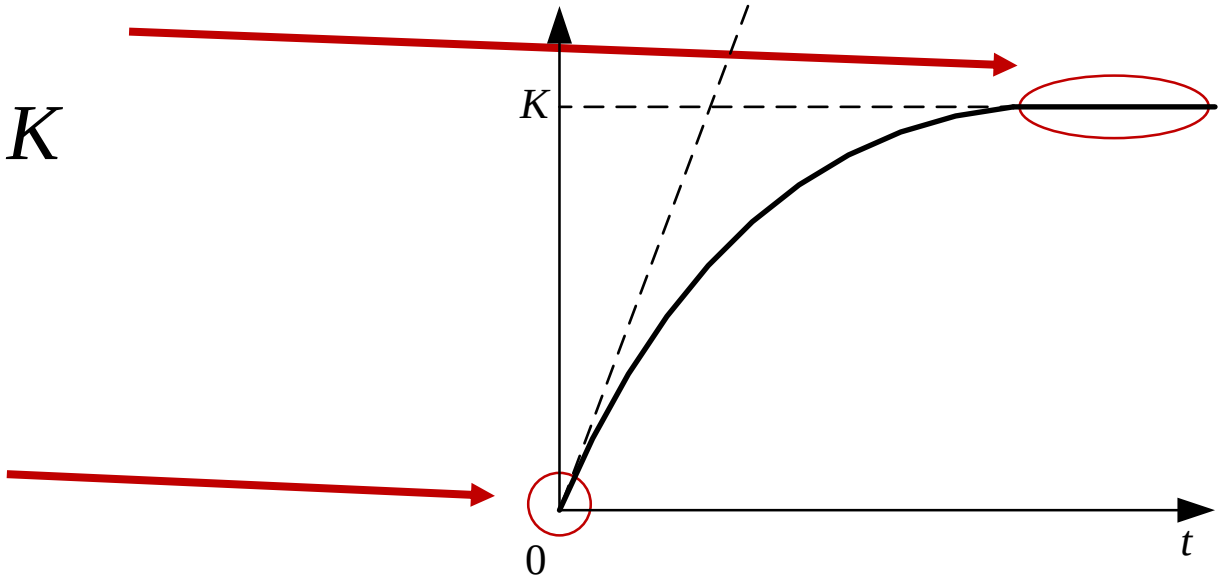
$$x(\infty)=K\left(1-e^{-\infty/\tau}\right)=K$$

$$x(\infty)=\lim_{s\to 0}sX(s)=\lim_{s\to 0}\frac{K}{\tau s+1}=K$$

- Initial Conditions

$$x(0)=K\left(1-e^{-0}\right)=0$$

$$x(0)=\lim_{s\to\infty}sX(s)=\lim_{s\to\infty}\frac{K}{\tau s+1}=0$$

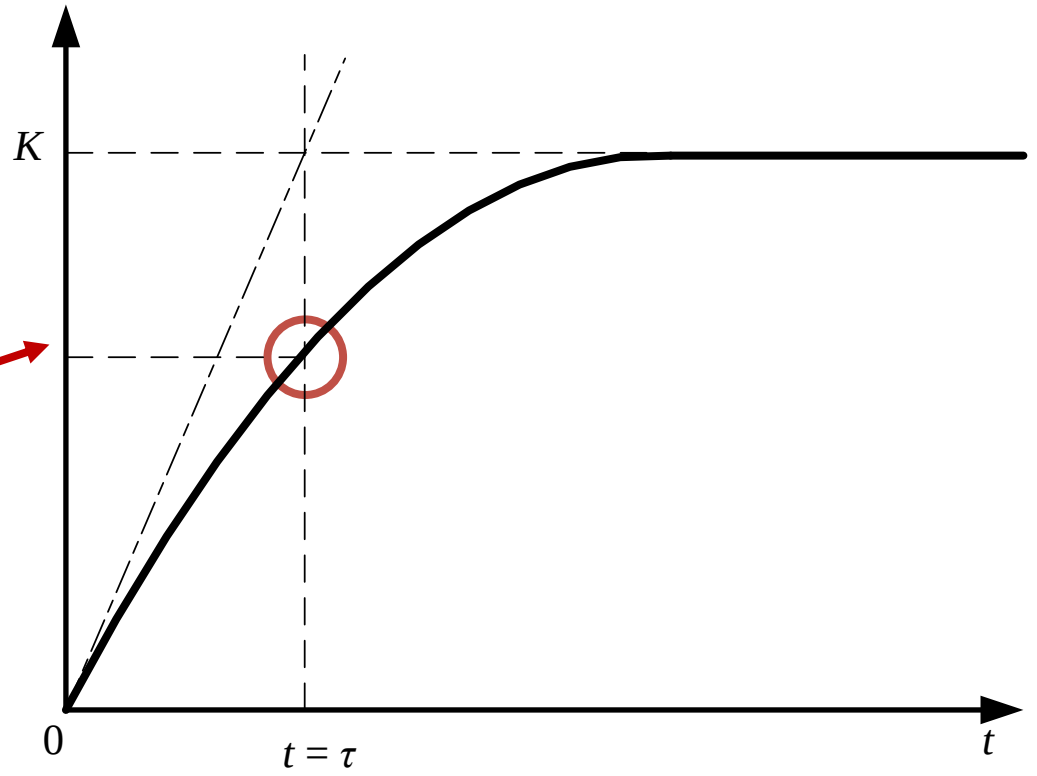


Response to a Unit Step - 2

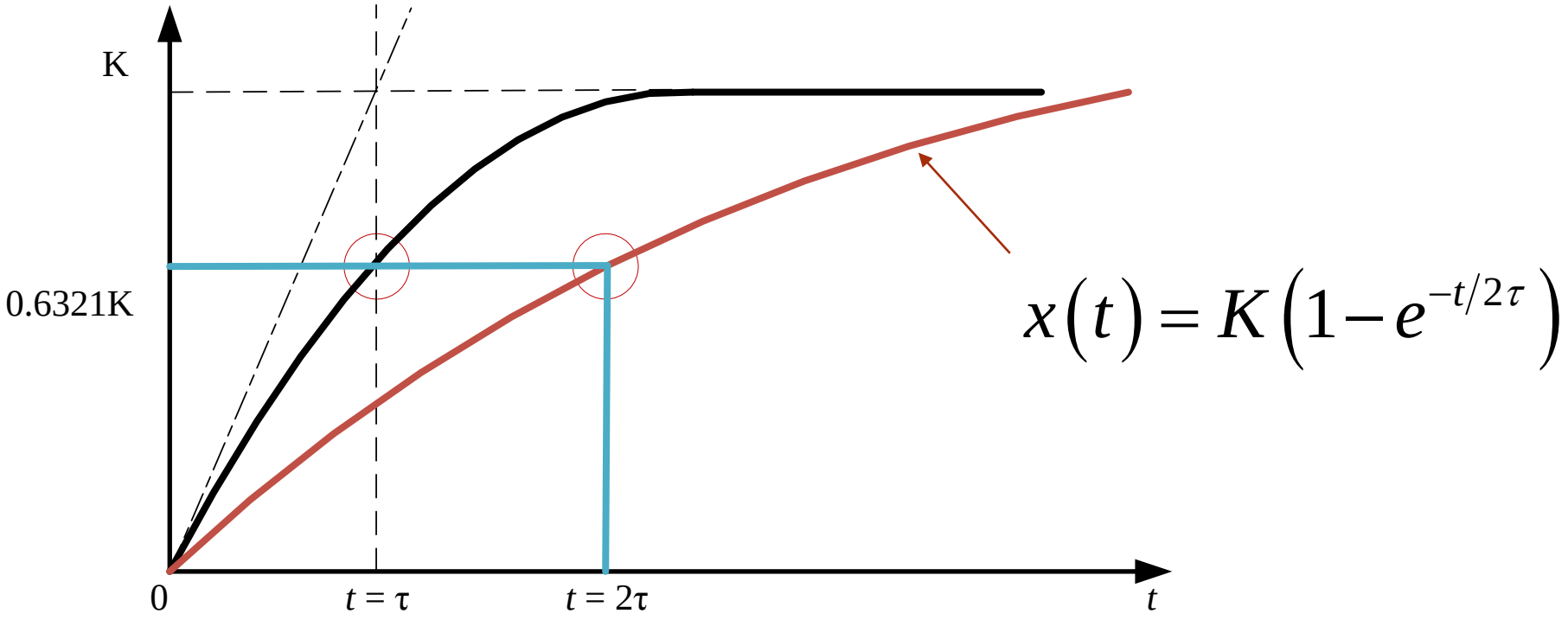
$$\tau \frac{dx}{dt} + x = Ku(t)$$

Time Constant

$$x(\tau) = K\left(1 - e^{-\tau/\tau}\right) = 0.632K$$



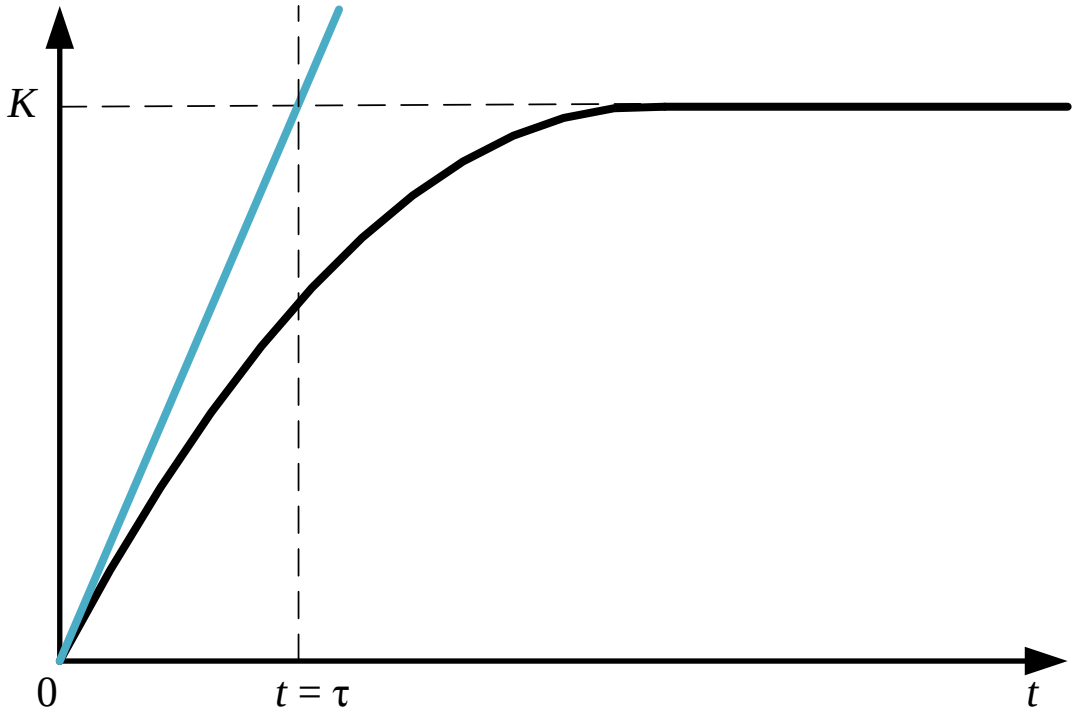
Response to a Unit Step - 3



Response to a Unit Step - 4

$$\tau \frac{dx}{dt} + x = Ku(t)$$

Initial speed of response



Response to a Unit Step - 5

- For a unit step input, the output gradually approaches the steady state gain of the system

$$x\left(t \to \infty\right) = K$$

Response to a Unit Step - 6

- To find the time constant from the step response graph, measure the time it takes for the output to reach 63% of its final value

- For a unit step input

$$x\left(t=\tau\right)=K\left(1-e^{-1}\right)=0.632K=0.632\ x\left(t\to\infty\right)$$

Example: Response to a Unit Step, solve $x(x=\tau)$

$$10\dot{x} + x = 1.5u(t)$$

Solution

$$10\dot{x} + x = 1.5u(t) \longrightarrow x = 1.5\left(1 - e^{-t/10}\right)$$

$$K = 1.5$$

$$x(t = \tau) = 0.632x(t \to \infty) = 0.632 \times 1.5 = 0.948$$

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1st Order - Frequency Response - 1

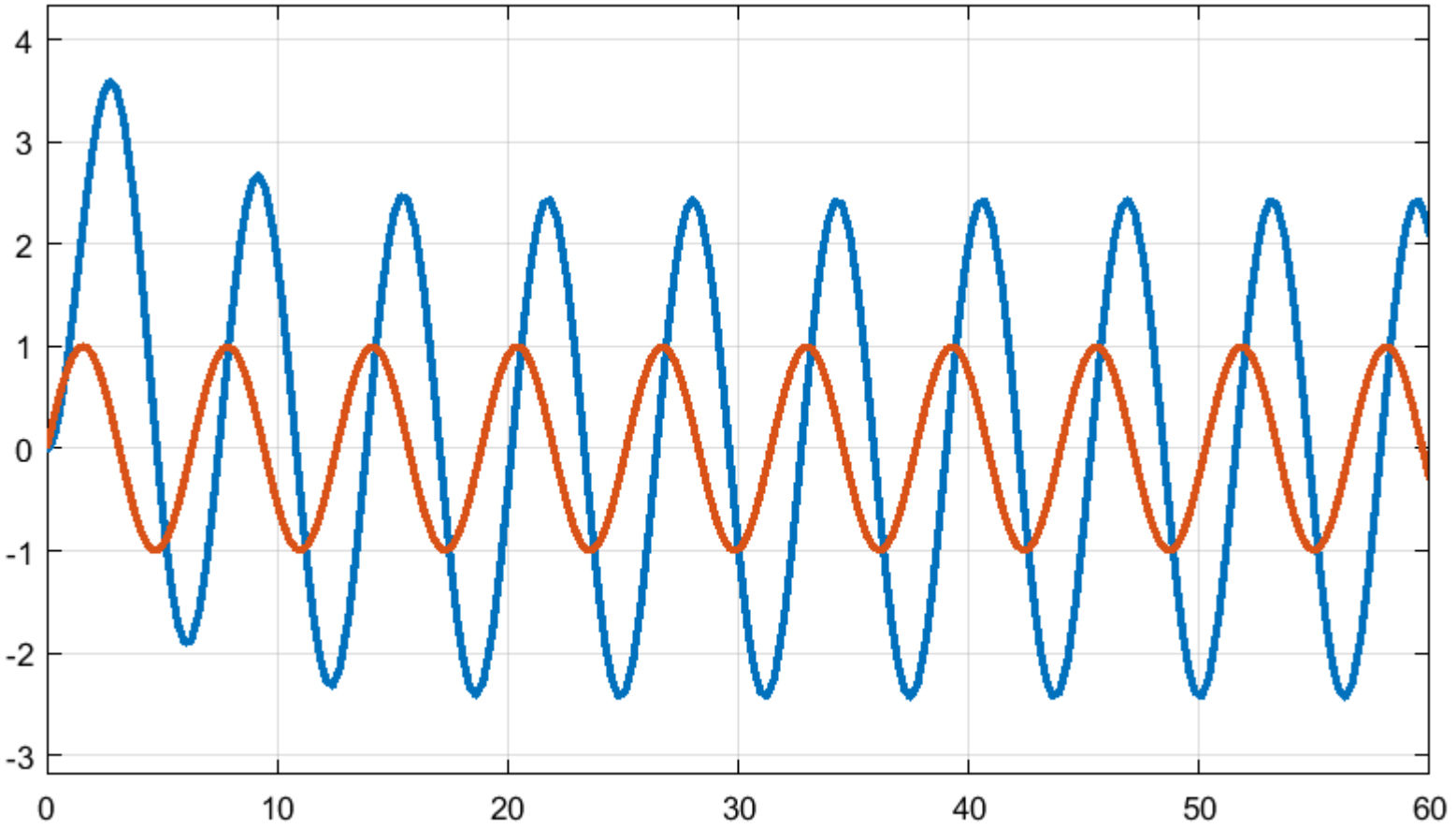
$$\tau \frac{dx}{dt} + x = K \cdot A\cos\left(\omega t + \theta\right)$$

The steady state response has the same frequency as the input

but a different amplitude and phase.

The amplitude and phase are functions of frequency.

1st Order - Frequency Response - 2



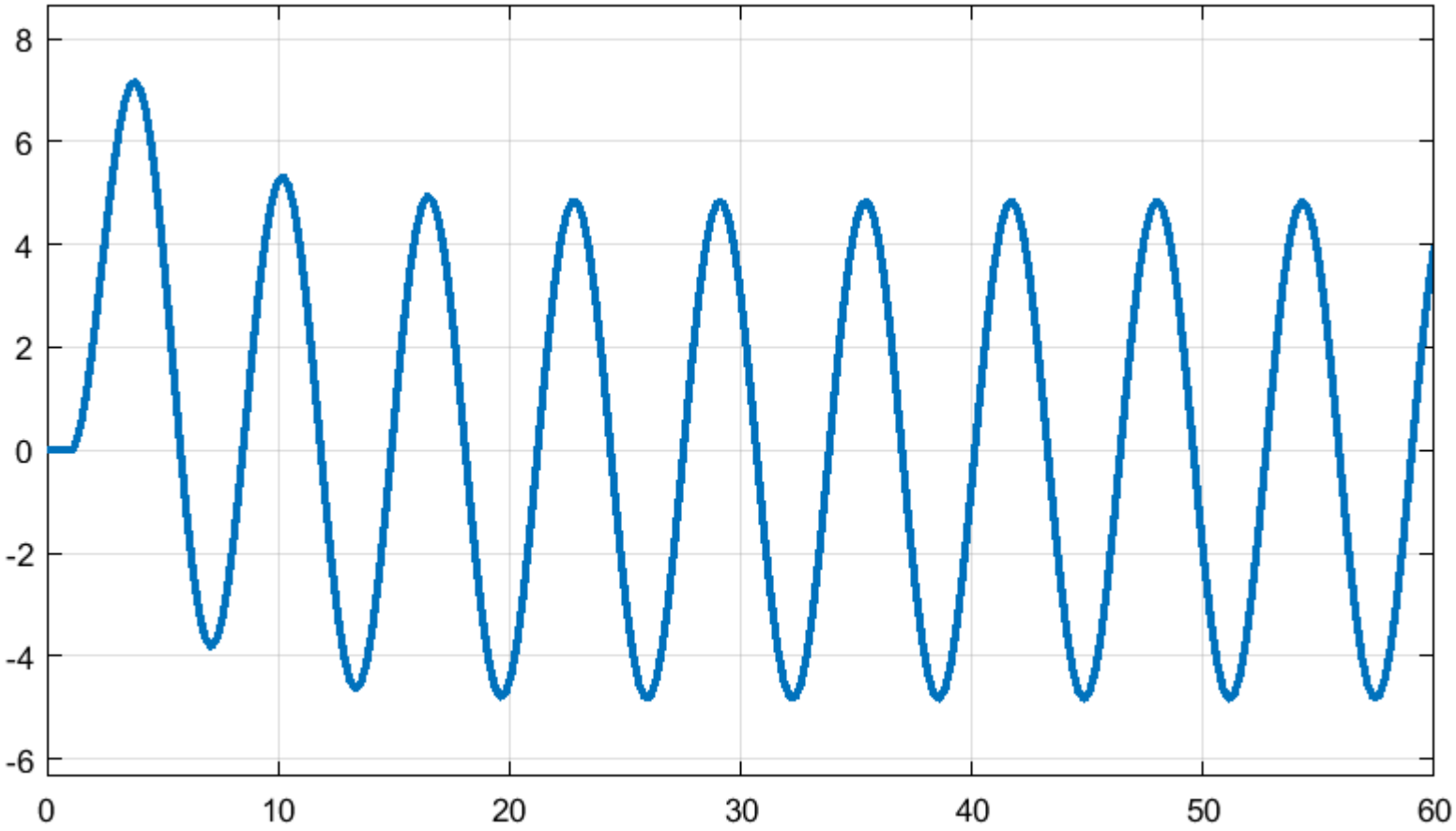
1st Order – Frequency Response – 3

$$\tau \frac{dx}{dt} + x = K \cdot A\cos\left(\omega t + \theta\right)$$

- It can be shown that the steady state response has the same frequency as the input but a different amplitude and phase.
- The amplitude and phase are functions of frequency.

$$X\left(s\right) = \frac{KA\left(s\cos\theta - \omega\sin\theta\right)}{\left(s^2 + \omega^2\right)\left(\tau s + 1\right)}$$

1st Order – Frequency Response – 4



Class question 1

Find the final value of the system corresponding to $Y(s)=\dfrac{3(s+2)}{s(s^2+2s+10)}$

Class question 2

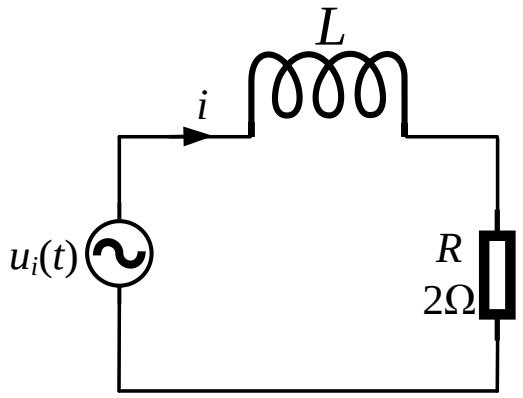
Find the solution to $\ddot{y}(t)+5\dot{y}(t)+4y(t)=u(t)$ $y(0)=0,\ \dot{y}(0)=0$

Using partial-fraction expansion

$$u(t)=\begin{cases}0 & t<0\\ 2e^{-2t} & t>=0\end{cases}$$

Class question 3

In the right figure， $u_i(t)=e^{-3t}$, determine the current $i(t)$.



Class question 1

Find the final value of the system corresponding to

$$Y(s) = \frac{3(s+2)}{s(s^2+2s+10)}$$

Solution

Applying the Final Value Theorem, we obtain

$$y(\infty) = sY(s)\Big|_{s=0} = \frac{3\times 2}{10} = 0.6$$

Thus, $y(t)$ will settle to a constant value of 0.6.

Class question 2

Find the solution to $\ddot{y}(t)+5\dot{y}(t)+4y(t)=u(t)$ $y(0)=0,\ \dot{y}(0)=0$

$$u(t)=\begin{cases}0 & t<0\\ 2e^{-2t} & t>=0\end{cases}$$

using partial-fraction expansion

Solution

- Taking the Laplace transform of both sides, we get $$s^2Y(s)+5sY(s)+4Y(s)=\frac{2}{s+2}$$

- Solving for $Y(s)$ yields $$Y(s)=\frac{2}{(s+2)(s+1)(s+4)}$$

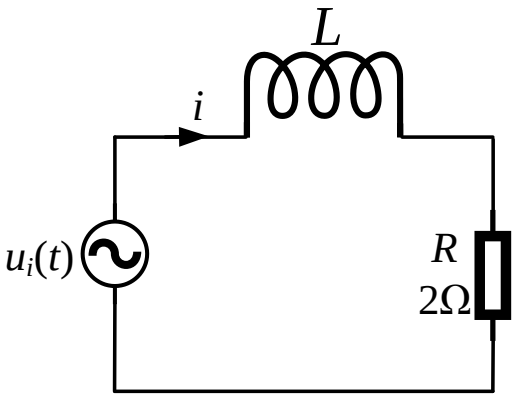
- The partial-fraction expansion $$Y(s)=-\frac{1}{s+2}+\frac{2/3}{s+1}+\frac{1/3}{s+4}$$

- Therefore, the time function is given by

$$y(t)=-e^{-2t}+\frac{2}{3}e^{-t}+\frac{1}{3}e^{-4t}$$

Class question 3

In the right figure， $u_i(t)=e^{-3t}$, determine the current $i(t)$.



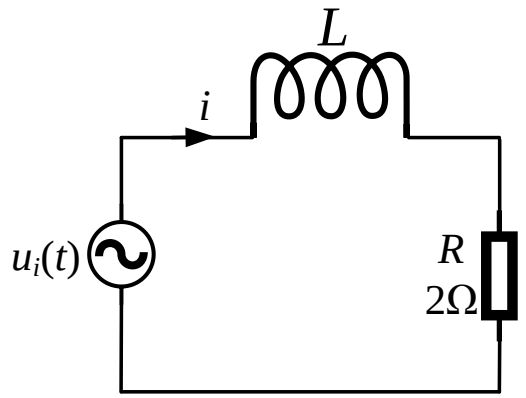
Solution

$$L\frac{di(t)}{dt}+Ri(t)=u_i(t)$$

$$sI+2I=\frac{1}{s+3}$$

$$I=\frac{1}{(s+3)(s+2)}=\frac{1}{s+2}-\frac{1}{s+3}$$

$$i(t)=e^{-2t}-e^{-3t}$$



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2nd Order - Solution - 1

- 2nd order

$$\frac{d^2x}{dt^2} + 2\zeta\omega_n \frac{dx}{dt} + \omega_n^2 x = Ku(t)$$

- Laplace transformation

$$s^2X + 2\zeta\omega_n sX + \omega_n^2 X = K/s$$

- So, we get

$$X = \frac{K}{s\left(s^2 + 2\zeta\omega_n s + \omega_n^2\right)}$$

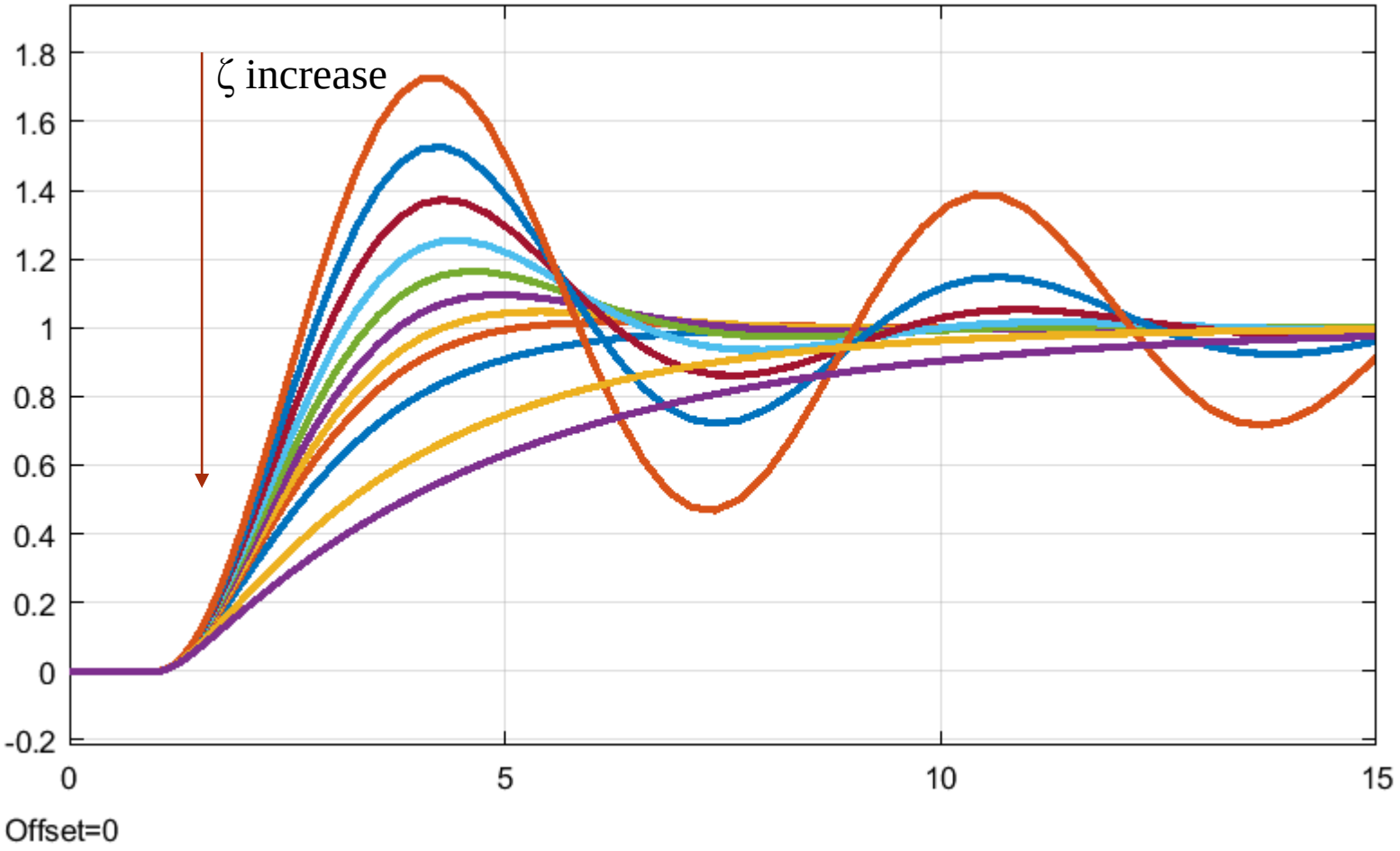
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2nd Order - Time response - 1

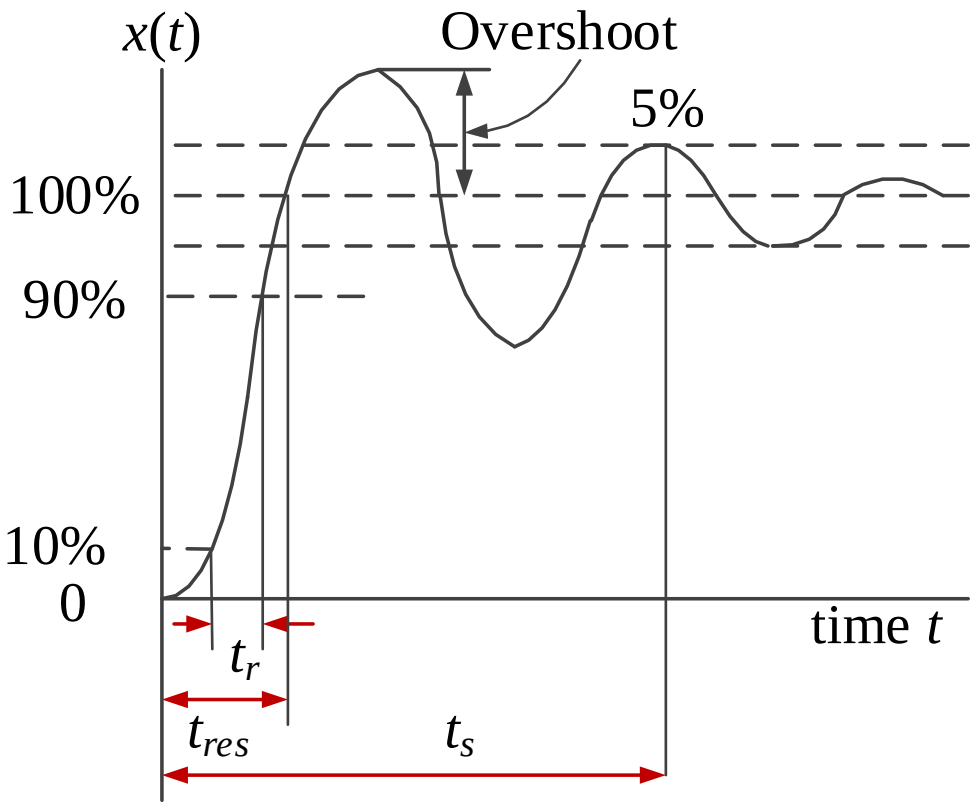
Response to a

Unit Step



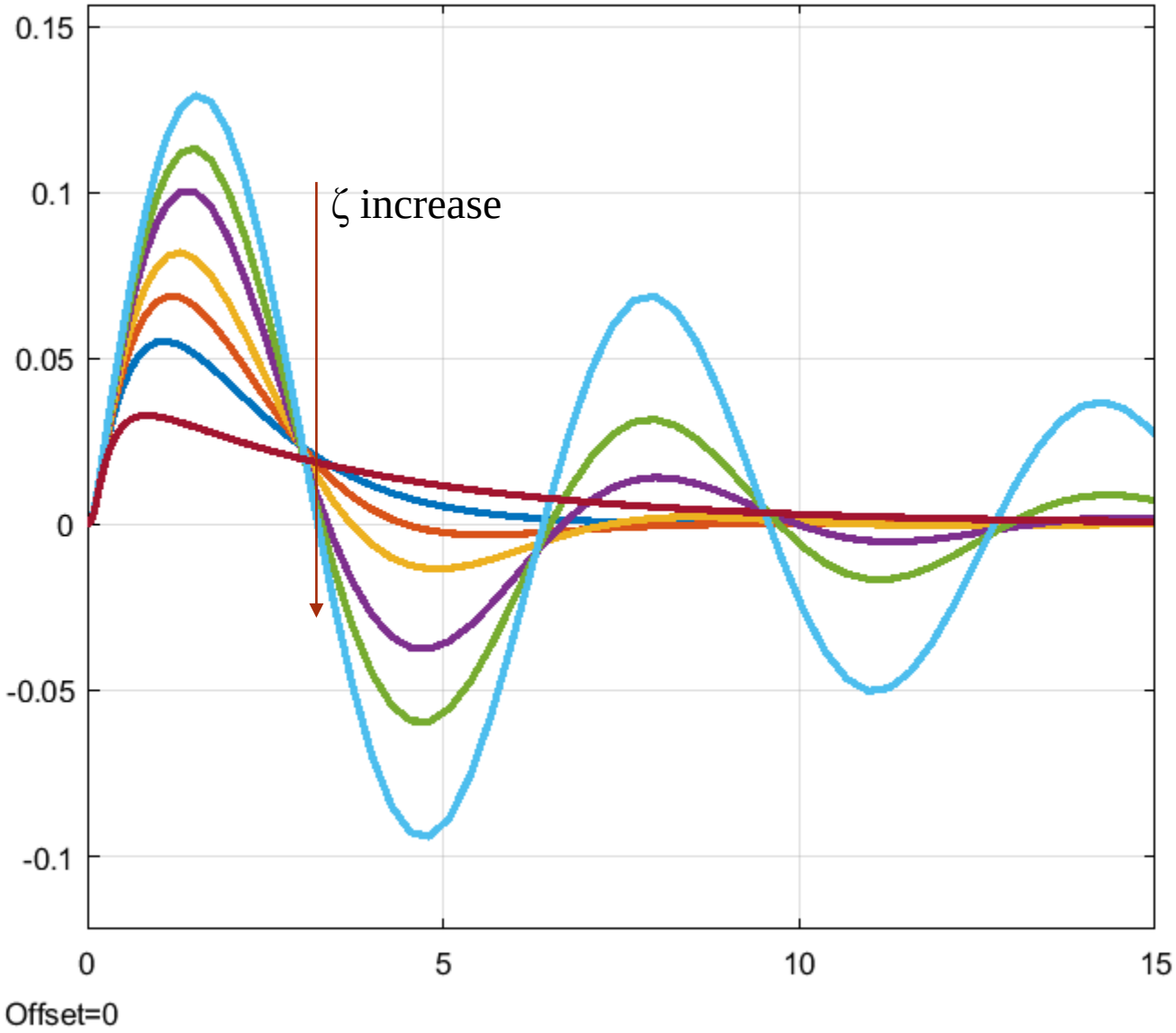
2nd Order - Time response - 2

- Response time t_{res}
- Rise time t_r
- Settling time t_s
- Overshoot



2nd Order - Time response - 3

- Response to a Unit Impulse
- We can obtain the solution directly by finding the derivative of the unit step response
- Identical to the transient response with initial conditions



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2nd Order - Frequency - 1

$$X = \frac{K}{s\left(s^2 + 2\zeta\omega_n s + \omega_n{}^2\right)}$$

$$= \frac{K_1}{s} + \frac{K_2\left(s + \zeta\omega_n\right)}{\left(s + \zeta\omega_n\right)^2 + \left(\sqrt{1-\zeta^2}\,\omega_n\right)^2} + \frac{K_3\left(\sqrt{1-\zeta^2}\,\omega_n\right)}{\left(s + \zeta\omega_n\right)^2 + \left(\sqrt{1-\zeta^2}\,\omega_n\right)^2}$$

$$x = K_1 + K_2 e^{-\zeta\omega_n t}\cos\left(\sqrt{1-\zeta^2}\,\omega_n t\right) + K_3 e^{-\zeta\omega_n t}\sin\left(\sqrt{1-\zeta^2}\,\omega_n t\right)$$

Class question 1

$$x(t) = t\cos(at) \longrightarrow X(s) = ?$$

Solution

$$x(t) = t\cos(at)$$

$$X(s) = \int_0^\infty t\cos(at)e^{-st}dt = \int_0^\infty t\frac{e^{jat}+e^{-jat}}{2}e^{-st}dt$$

$$= \frac{1}{2}\left(\int_0^\infty te^{-(s-ja)t}dt + \int_0^\infty te^{-(s+ja)t}dt\right)$$

$$= \frac{s^2-a^2}{\left(s^2+a^2\right)^2}$$

Class question 2

$$X(s) = \frac{s}{s^2 + 2s + 5} \longrightarrow x(t) = ?$$

Solution

$$X(s)=\frac{s}{s^2+2s+5}=\frac{(s+1)-1}{(s+1)^2+2^2}$$

$$=\frac{(s+1)}{(s+1)^2+2^2}-\frac{1}{2}\frac{2}{(s+1)^2+2^2}$$

$$x(t)=e^{-t}\cos(2t)-0.5e^{-t}\sin(2t)$$

Summary

- The 1st and 2nd order differential equations is solved by Laplace
- Standard cases include impulse, step, ramp and harmonic response
- The impulse response is related to the transient behaviour
- The unit step is useful for identifying model coefficients, i.e. time constant, damping ratio and natural frequency
- The steady state response to a harmonic input has the same frequency, but different amplitude and phase
- polar plots – this is the frequency response